

CANCELLATION PREDICTION

PREPARED BY:
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DATA

Source: Kaggle – Hotel Booking Demand

Total observations: Over 119,000 hotel bookings

Our working dataset: 13 selected features

Types of hotels: City Hotel and Resort Hotel

Time period: Bookings collected from 2015 to 2017



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OBJECTIVE: PREDICT WHETHER A BOOKING WILL BE CANCELED OR NOT



Features

Output Variable: is_canceled

Categorical: hotel

Categorical: meal

Categorical: market segment

Categorical: deposit_type

Categorical: customer_type

Numerical: lead_time

Numerical: adults

Categorical: is_repeated_guest

Numerical: previous cancellations

Numerical: booking_changes

Numerical: adr

Numerical: total_of_special_requests

NUMERICAL VARIABLES



ANALYSIS

Variable	Mean	Std	Max Value	Min Value
lead_time	104.01	106.86	737	0
adults	1.86	0.58	55	0
previous cancellations	0.09	0.84	26	0
booking changes	0.22	0.65	21	0
adr (Avg Daily Rate)	101.83	50.54	5400	-6.38
total of special requests	0.57	0.79	5	0

CATEGORICAL VARIABLES

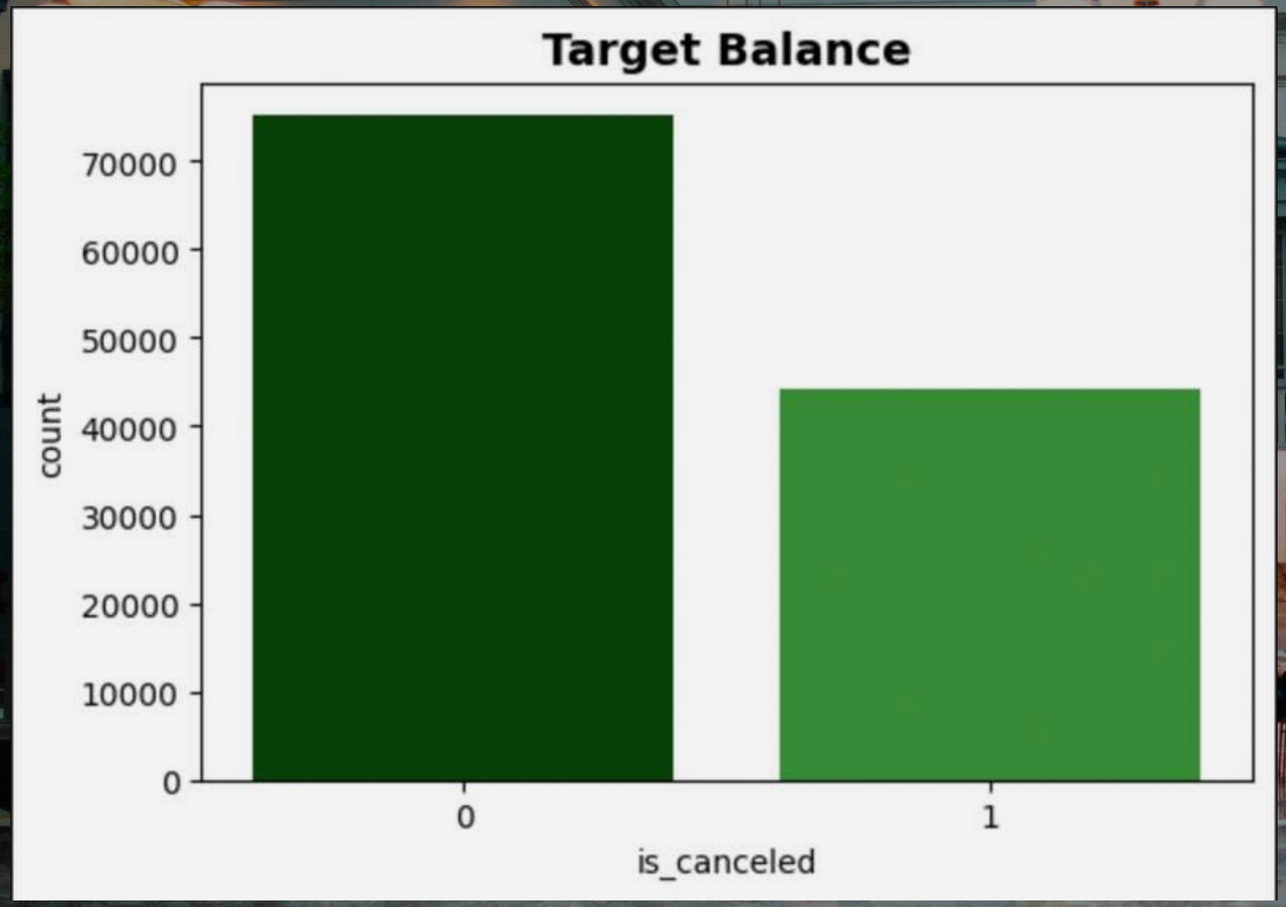


ANALYSIS

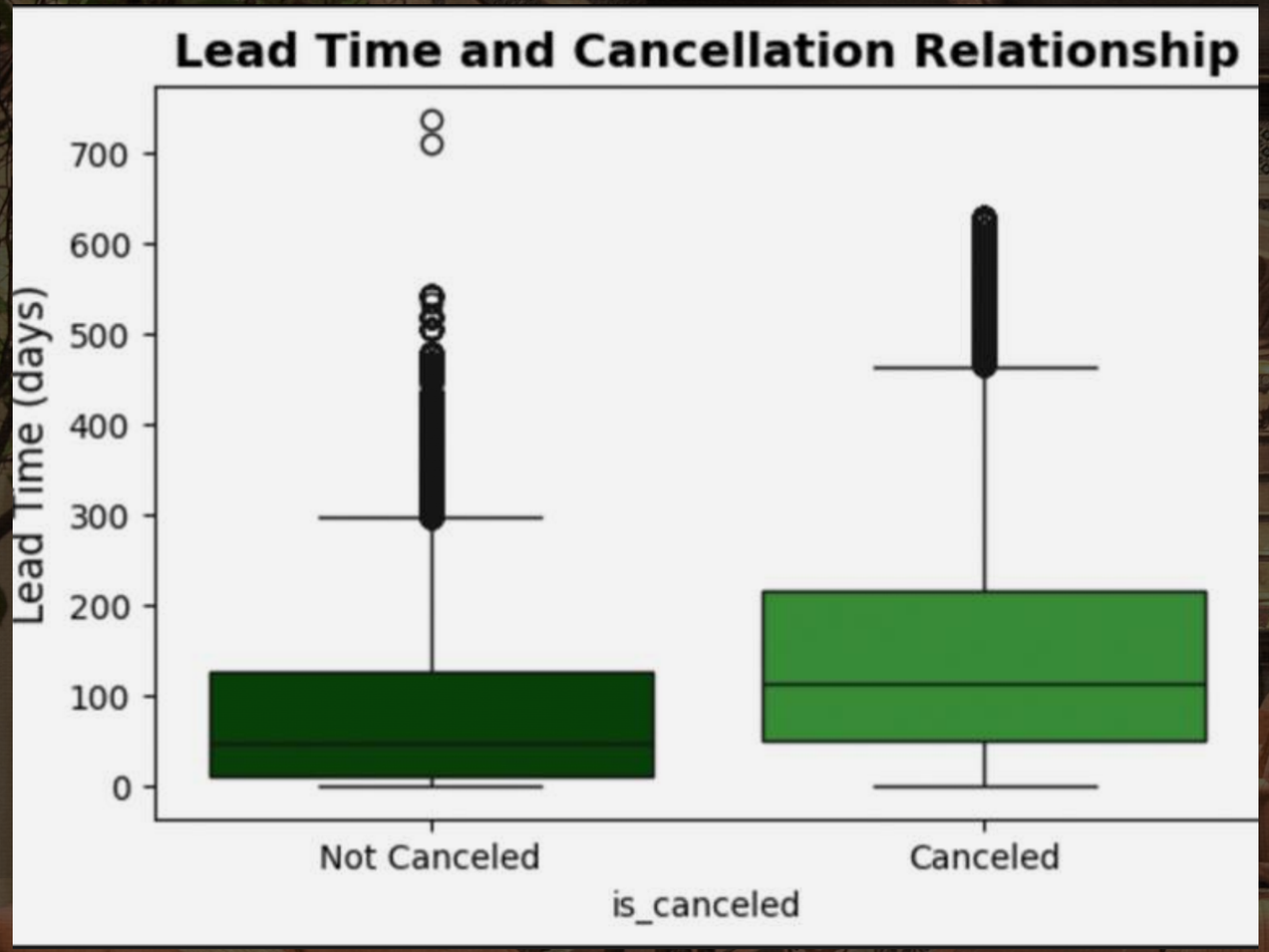
Variable	Unique Values	Most Frequent	Frequency	Notes
is_canceled	2	0 (Not canceled)	63	37% canceled
hotel	2	City Hotel	79,330	City > Resort
meal	5	BB (Bed & Breakfast)	92,310	BB dominates
market_segment	8	Online TA	56,477	OTA is main channel
deposit_type	3	No Deposit	104,641	Most bookings no deposit
customer_type	4	Transient	89,613	Mostly individual bookings
is repeated guest	2	0 (Not repeated)	96.8	Only 3.2% repeated guests

Visualization & Insights

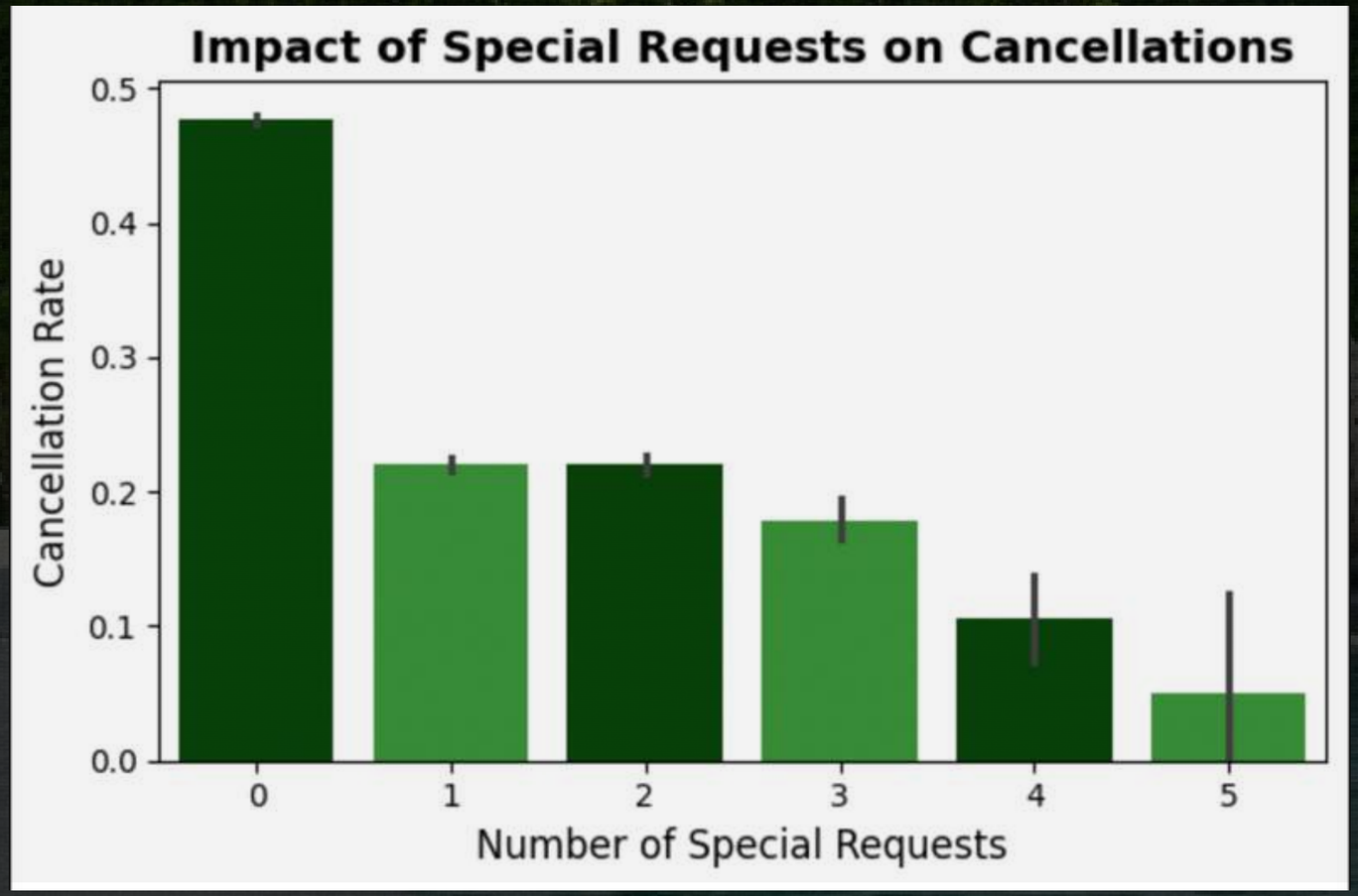




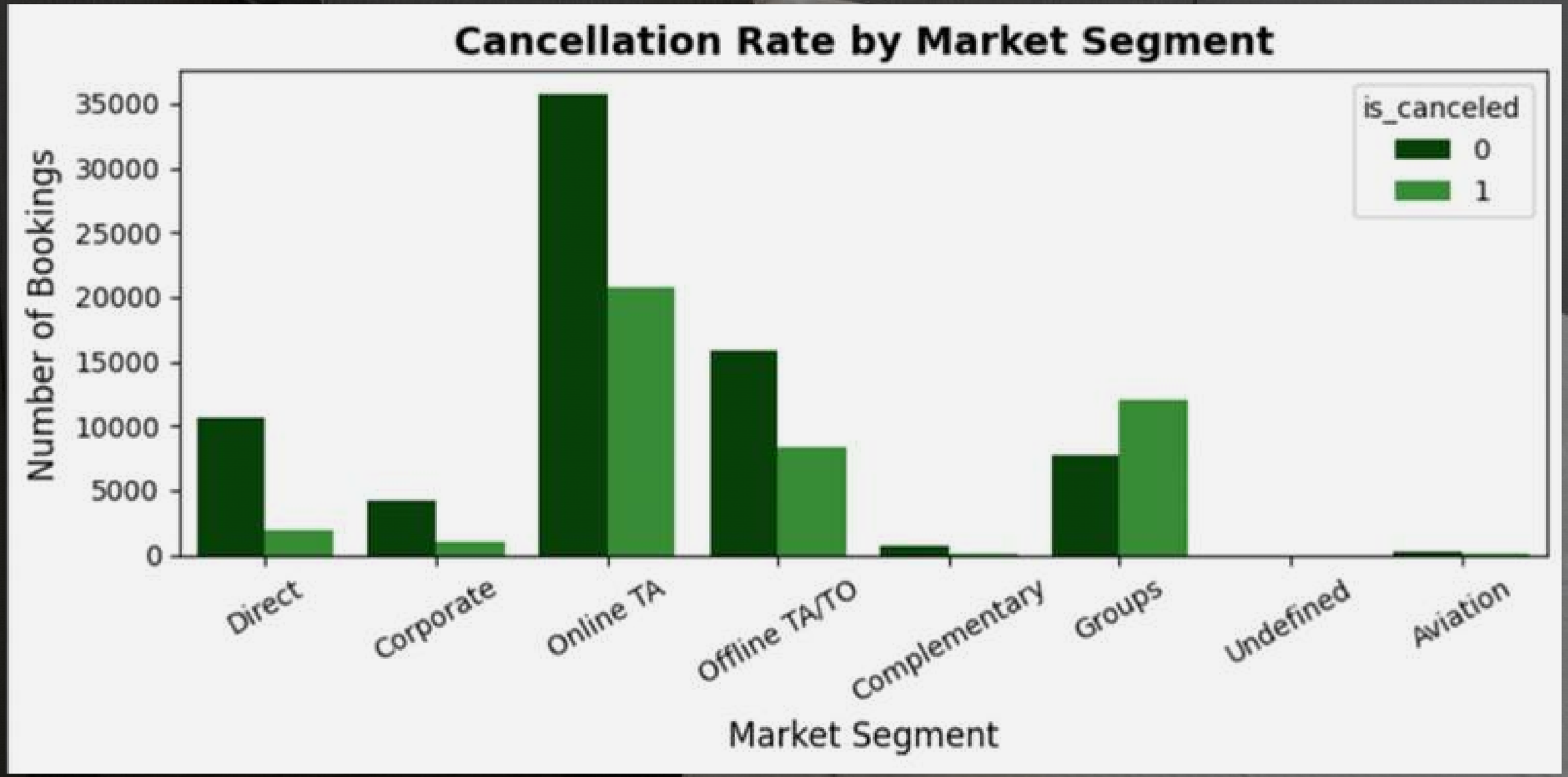
THE DATASET SHOWS THAT **37.04%** OF BOOKINGS ARE CANCELED.



Bookings made far in advance are **more likely** to be canceled



GUESTS WITH MORE SPECIAL REQUESTS ARE **LESS LIKELY** TO CANCEL


Cancellation Rate by Market Segment
Number of Bookings
35000
30000
25000
20000
15000
10000
5000
0
is_canceled
0
1
Direct
Corporate
Online TA
Offline TA/TO
Complementary
Groups
Undefined
Aviation
Market Segment

DATA PREPARATION

NO MISSING VALUES DETECTED, SO **NO** IMPUTATION NEEDED

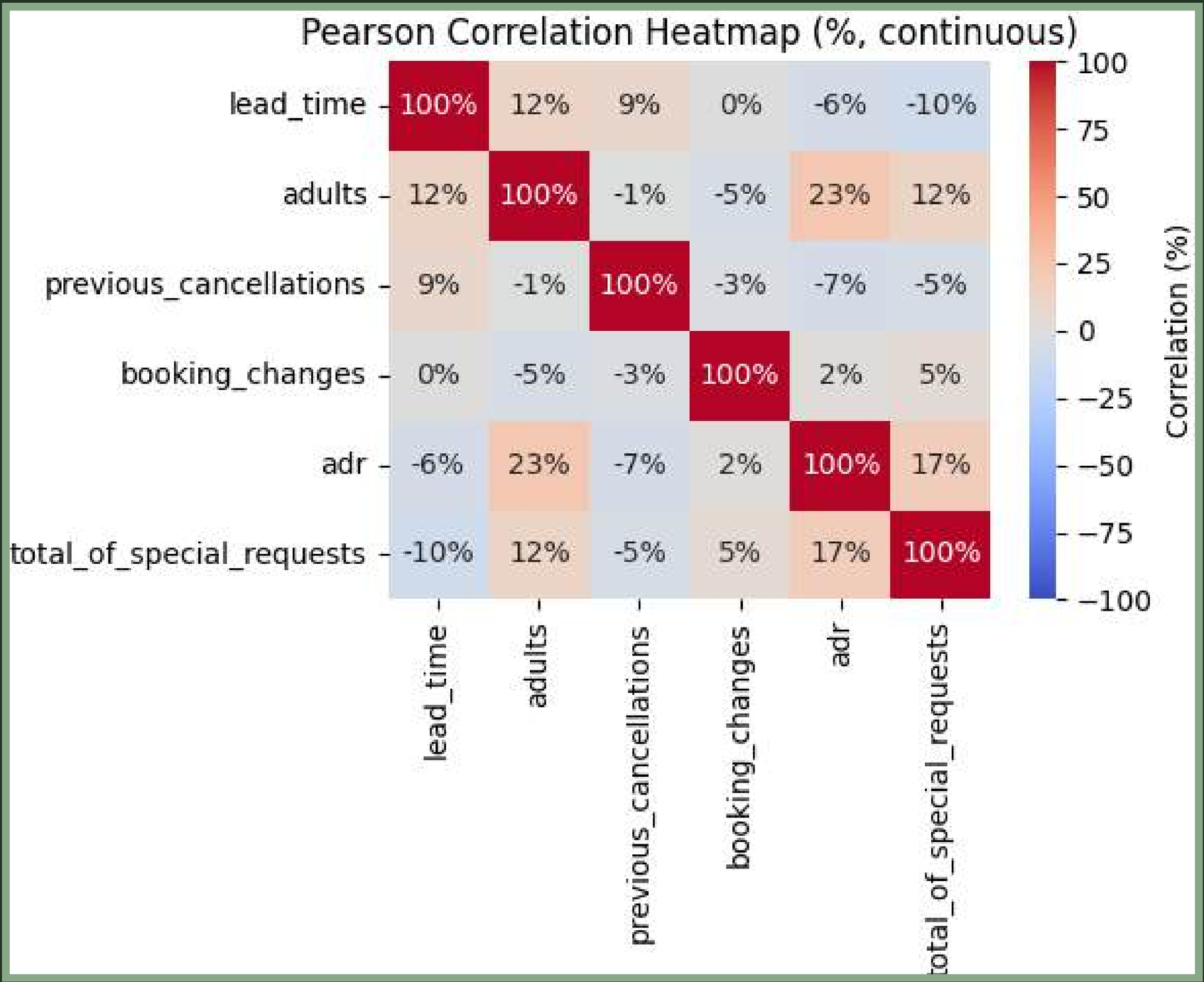
ENCODED CATEGORICAL FEATURES INTO NUMERICAL FORMAT

SPLIT DATASET INTO FEATURES **(X)** AND TARGET **(Y)** – TARGET: **IS_CANCELED**

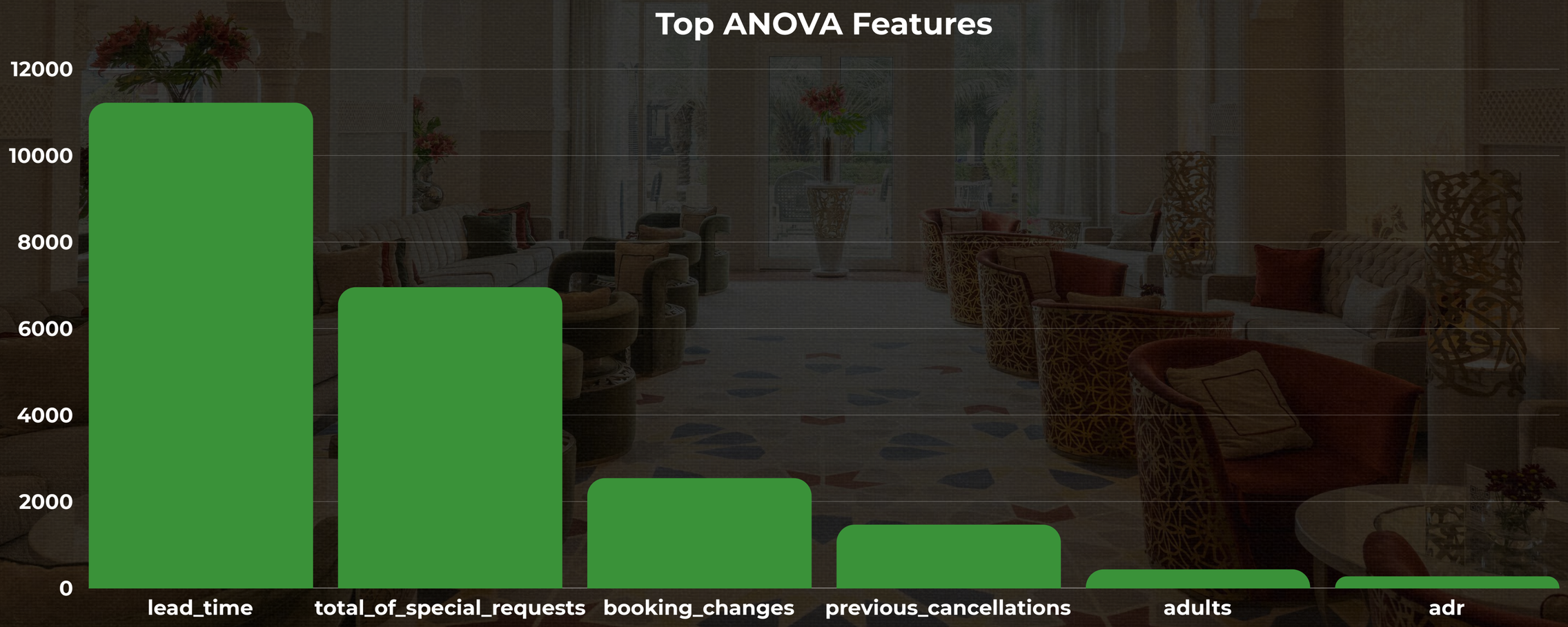


FEATURE SELECTION PROCESS

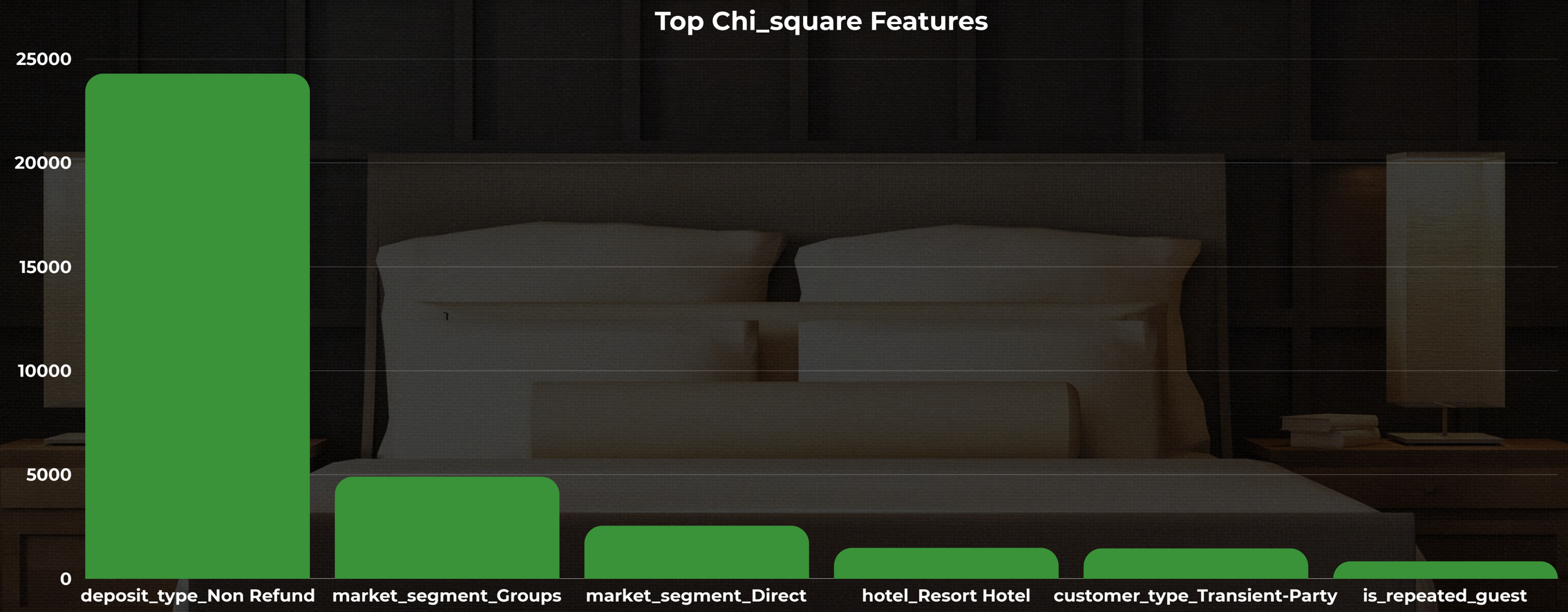
1. **Filter Method:** Low Variance
2. **Settings:** k_anova, k_chi2, k_rfe
3. **Pearson heatmap**
4. **Statistical Tests:** ANOVA and Chi-square
5. **Wrapper Method:** RFE

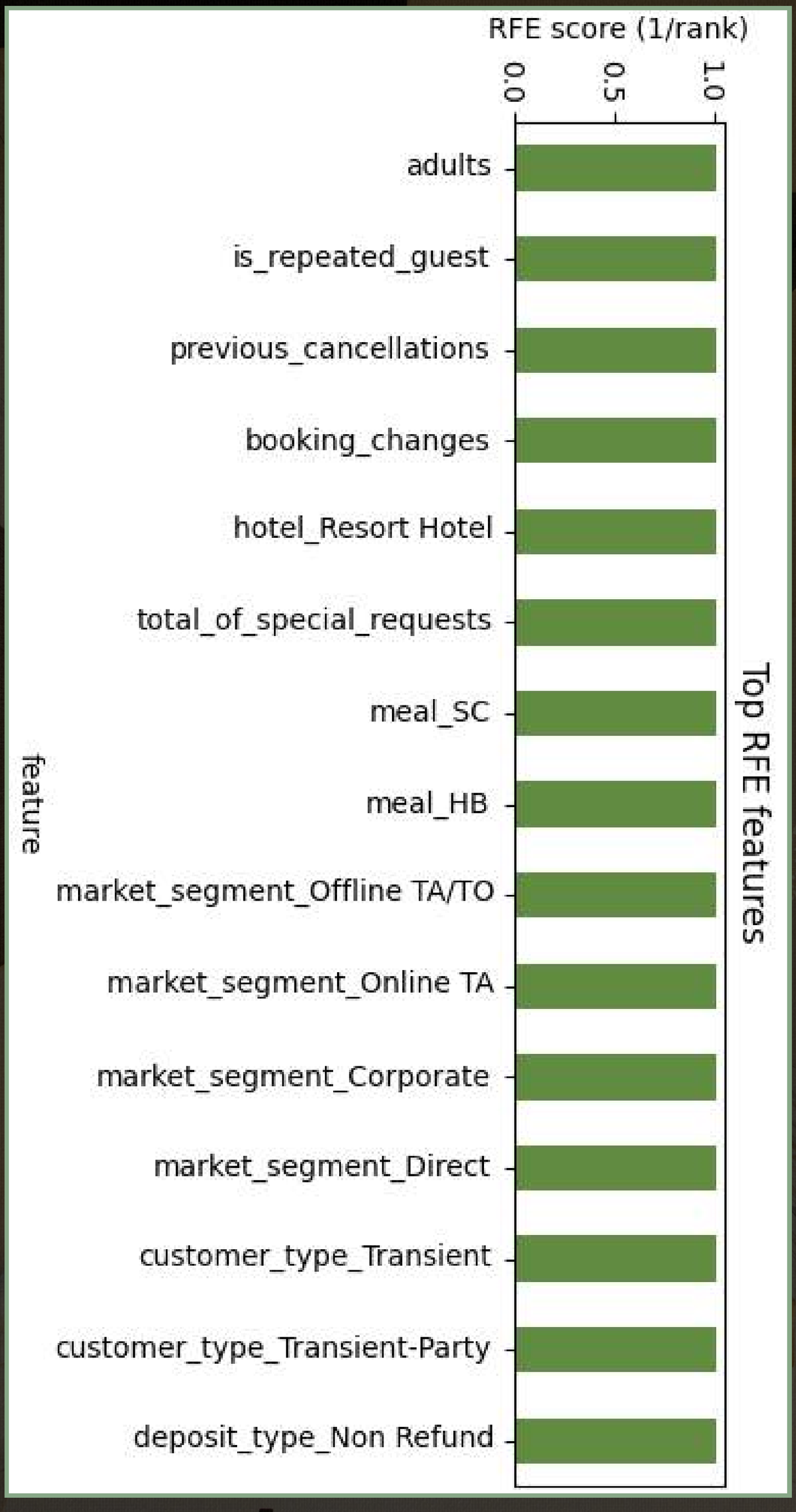

Pearson Correlation Heatmap (%, continuous)
lead_time
adults
previous_cancellations
booking_changes
adr
total_of_special_requests
100%
12%
9%
0%
-6%
-10%
12%
100%
-1%
-5%
23%
12%
9%
-1%
100%
-3%
-7%
-5%
0%
-5%
-3%
100%
2%
5%
-6%
23%
-7%
2%
100%
17%
-10%
12%
-5%
5%
17%
100%
Correlation (%)
100
75
50
25
0
−25
−50
−75
−100

ANOVA


Top ANOVA Features
12000
10000
8000
6000
4000
2000
0
lead_time
total_of_special_requests
booking_changes
previous_cancellations
adults
adr

Chi-square


Top Chi_square Features
25000
20000
15000
10000
5000
0
deposit_type_Non Refund
market_segment_Groups
market_segment_Direct
hotel_Resort Hotel
customer_type_Transient-Party
is_repeated_guest


RFE score (1/rank)
0.0
0.5
1.0
adults
is_repeated_guest
previous_cancellations
booking_changes
hotel_Resort Hotel
total_of_special_requests
meal_SC
meal_HB
market_segment_Offline TA/TO
market_segment_Online TA
market_segment_Corporate
market_segment_Direct
customer_type_Transient
customer_type_Transient-Party
deposit_type_Non Refund
feature
Top RFE features

Final Selection

Output Variable: is_canceled	market segment Groups	market segment Offline TA/TO	market segment Direct
market segment Corporate	meal_HB	lead_time	adults
is_repeated_guest	meal_SC	booking_changes	adr
deposit_type_Non Refund	hotel_Resort Hotel	total of special requests	customer type Transient
previous cancellations	customer type Transient Party	market segment Online TA	

MODEL'S SUMMARY



Model	Description	Hyperparameters
Linear SVM	**Linear classifier separating data with a hyperplane**	**C = [0.1, 1, 10], loss = ['squared_hinge', 'hinge']**
Decision Tree	Tree-based splitting by feature values	max_depth = [None, 5, 10, 20], min_samples_split = [2, 5, 10], min_samples_leaf = [1, 2, 4]
Random Forest	**Ensemble of decision trees using bagging**	**n_estimators = [100], max_depth = [None, 10], min_samples_split = [2, 5], min_samples_leaf = [1, 2]**
Naive Bayes	Probabilistic model assuming feature independence	var_smoothing = np.logspace(-10, -7, 4)
Logistic Regression	Linear model estimating class probabilities	C = [0.1, 1, 10], penalty = 'l2', solver = 'lbfgs', class_weight = [None, 'balanced']
Gradient Boosting	**Sequentially builds weak trees to reduce errors**	**n_estimators = [100], learning_rate = [0.1], max_depth = [3]**
ANN	**Multi-layer neural network with non-linear activation**	**hidden_layer_sizes = [(64, 32)], alpha = [1e-4], batch_size = [128], learning_rate_init = [1e-3]**

MODEL'S



PERFORMANCE

Model	Accuracy	F1-Score
Linear SVM	0,793	0.651
Decision Tree	0.803	0.696
Random Forest	0.814	0.704
Naive Bayes	0.760	0.536
Logistic Regression	0.775	0.695
Gradient Boosting	0.811	0.702
ANN	0.815	0.716

CONFUSION MATRIX (ANN)

	Predicted: No Cancel	Predicted: Cancel
Actual: No Cancel	69543	5623
Actual: Cancel	16436	27788

- **69,543** bookings were **correctly predicted** as **not canceled**
- **27,788** bookings that were actually **canceled** were **correctly predicted** by the model
- **16,436 canceled** bookings were **not detected** by the model
- **5,623** bookings were **incorrectly predicted** as **canceled**

CONFUSION MATRIX (RANDOM FOREST)

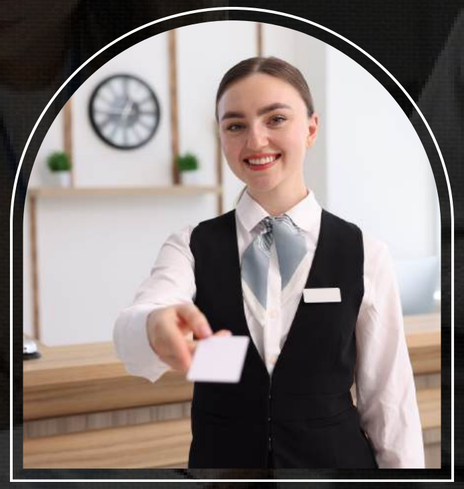
	Predicted: No Cancel	Predicted: Cancel
Actual: No Cancel	**70631**	**4535**
Actual: Cancel	**17721**	**26503**

- The model **correctly predicts** most non-cancellations
- It accurately identifies **26,503 real cancellations**
- However, **17,721** cancellations were **missed**
- **4,535** bookings were **wrongly predicted** as cancellations

SMART HOTEL STRATEGIES FROM MODEL INSIGHTS



Promote non-refundable deposit options to reduce cancellations



Monitor high lead-time bookings
➜ early reservations = higher risk

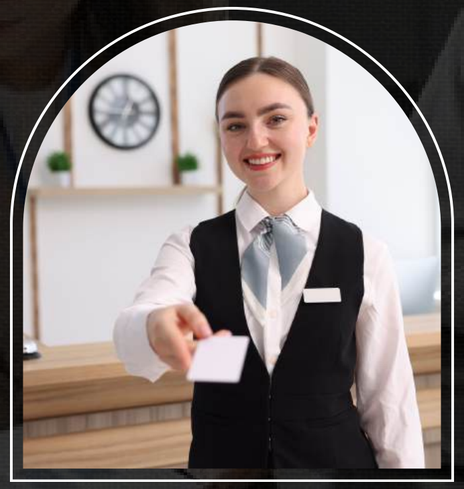


Prioritize guests with **special requests**

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Prioritize guests with **special requests**

Integrate ANN predictions into a Power BI dashboard for real-time decision-making